

# Automating the Development and Management of Intelligent Software Systems

DRAFT

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# 1 Introduction

Software systems are an essential part of modern society and when they fail to function as intended, the implications can be enormous. It is therefore vital that systems are built to minimize the possibility of failure and that, when failures do occur, effective diagnostic processes expedite the systems recovery and prevent the recurrence of failures.

To address this need, this document presents a framework called the Design Learning Platform (DLP) that automates the development and management of intelligent software systems.

## 2 The Design

### 2.1 Meaning of the World

### 2.2 Narrative of the World

### 2.3 Reasoning about the World

### 2.4 Shared Meaning

### 2.5 Semantic Invalidity

## 3 Computable Semantic Model

### 3.1 Implementing a computable semantic model

### 3.2 Computable Semantic Primitives

### 3.3 Automated Instrumentation

## 4 Semantic Toolkit

### 4.1 Automated Development

## 5 Semantic Reasoning Engine

### 5.1 Meaning of Failures

### 5.2 Design Abstraction Language

### 5.3 Types of Reasoning

#### 5.3.1 Reasoning about Internal Consistency

#### 5.3.2 Reasoning about Reality

### 5.4 Practical Considerations for Reasoning about Reality

# Design Learning Platform

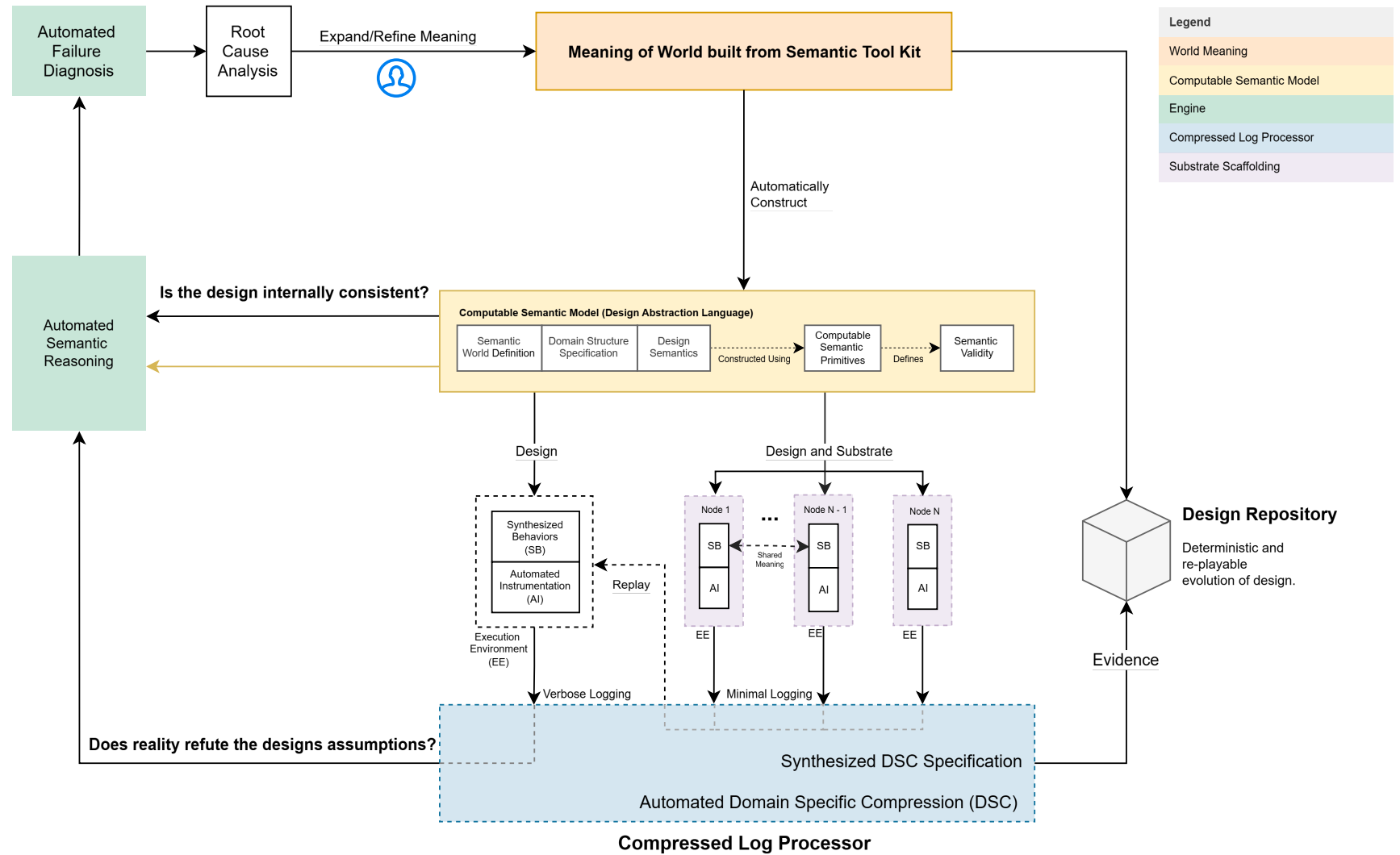


Figure 1: The Design Learning Platform

## 6 Domain Specific Compression

### 6.1 Compressed Log Processor

## 7 Design Learning Platform

## 8 Deterministic Intelligence

## **9 Additional Thoughts**

### **9.1 Automated Orchestration**

### **9.2 Securing Interaction**

### **9.3 Automated Documentation**

### **9.4 Machine Driven Engineering**

## 10 Conclusion

## **11 Extended Applications**

### **11.1 Universal Design Language**

### **11.2 Digital Twins**

### **11.3 Modelling Knowledge**

### **11.4 DCS for Archiving using Custom Specification**